Geographic Patterns in Diabetes and Related Risk Factors Across U.S. Counties

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HIT 723 Public Health Informatics

Final Part 3: Data Visualization and Explanation

1. INTRODUCTION

Diabetes is an important public health concern in the United States because it affects population health and can contribute to serious long term complications. According to the Centers for Disease Control and Prevention (CDC, 2024a), type 2 diabetes accounts for approximately 90%–95% of diabetes cases in the United States. Several risk factors, including overweight and obesity, physical inactivity, and family history, are associated with the development of type 2 diabetes. Understanding how diabetes prevalence varies across geographic areas can help public health professionals identify communities that may benefit from additional prevention, education, and healthcare resources.

The purpose of this project was to explore the geographic distribution of estimated diabetes prevalence across U.S. counties and examine its relationship with obesity prevalence. Using county level data from the Centers for Disease Control and Prevention (CDC) PLACES 2025 dataset, I developed three visualizations in Tableau Public: a geographic map of diabetes prevalence, a bar chart identifying the ten counties with the highest estimated diabetes prevalence, and a scatter plot examining the relationship between diabetes and obesity prevalence.

Together, these visualizations provide a way to explore geographic differences and potential relationships between chronic disease indicators. They also demonstrate how data visualization can support public health surveillance and health planning.

2. DATA SOURCE AND VARIABLES

The data used in this project came from the CDC's PLACES (Local Data for Better Health) 2025 release. PLACES provides model based estimates of chronic disease and other health-related measures at different geographic levels, including counties, cities, census tracts, and ZIP Code Tabulation Areas (CDC, 2025a). The 2025 PLACES county-level dataset uses information from the 2023 Behavioral Risk Factor Surveillance System (BRFSS), U.S. Census Bureau population estimates, and American Community Survey data. These sources support the development of consistent, model-based estimates of health indicators across geographic areas (Centers for Disease Control and Prevention [CDC], 2025c). The BRFSS is an ongoing

health survey that collects information about chronic health conditions, health-related behaviors, healthcare access, and preventive practices among adults (CDC, 2024b).

The dataset used for this analysis was the county level file named *PLACES_2025_diabetes_county_visualization_for_tableau.csv*. The Tableau data source contained 2,957 rows and 14 fields. The analysis focused on the 2023 estimates and selected variables relevant to diabetes and obesity.

The primary variables used were:

- **Year:** Identifies the reporting year of the estimate.
- **State:** Identifies the state in which the county is located.
- **County:** Identifies the county represented by each observation.
- **Diabetes Prevalence:** Estimated percentage of adults with diagnosed diabetes.
- **Obesity Prevalence:** Estimated percentage of adults with obesity.
- **County and State:** A combined geographic field used to identify individual counties in the visualizations.

The CDC PLACES estimates are generated using small-area estimation methods. These estimates use information from the Behavioral Risk Factor Surveillance System (BRFSS), U.S. Census population data, and American Community Survey data (CDC, 2025b). Therefore, the values represent modeled population-level estimates rather than direct measurements of every individual living in a county.

3. DATA VISUALIZATION AND FINDINGS

3.1 County-Level Diabetes Prevalence Map

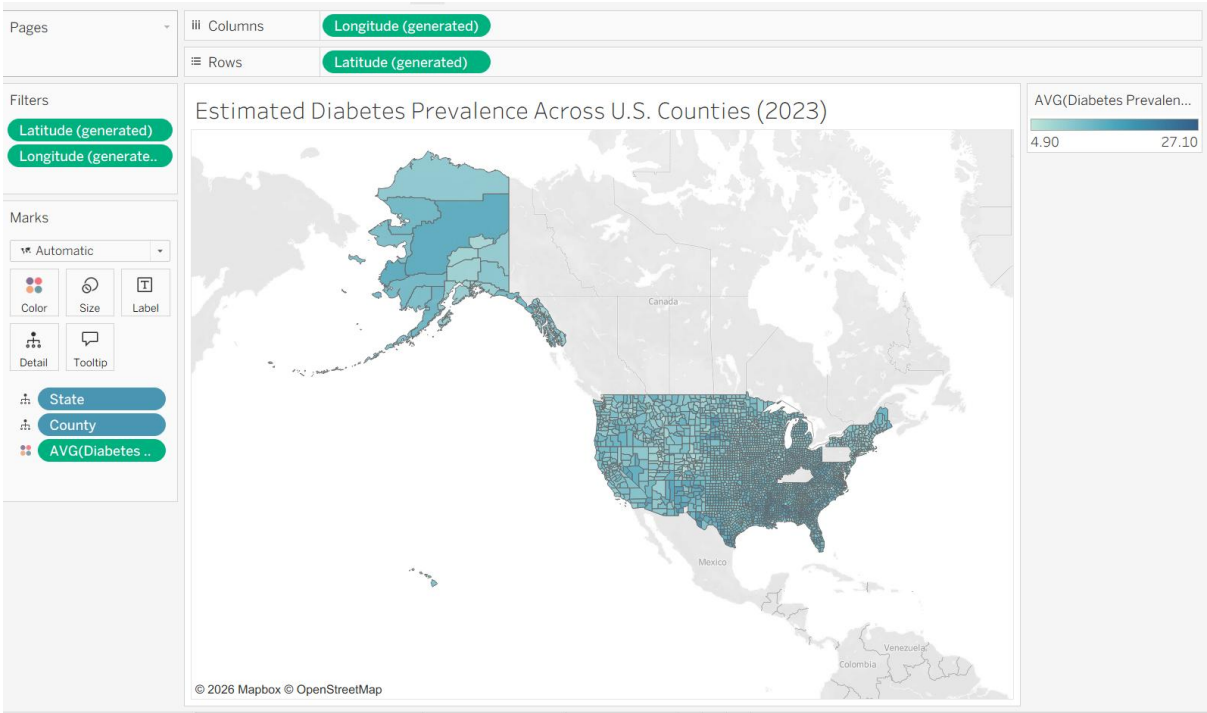


Figure 1. Estimated Diabetes Prevalence Across U.S. Counties (2023).

The first visualization is a geographic map displaying estimated diabetes prevalence across U.S. counties. I used Tableau's generated latitude and longitude fields to create the map and applied diabetes prevalence as the colour measure. The colour gradient represents differences in estimated prevalence across counties, with lighter and darker shades indicating different levels.

The map shows that diabetes prevalence is not geographically uniform. There are visible differences among counties across the United States, suggesting that some communities experience a greater estimated burden of diabetes than others. The displayed estimates range from approximately 4.9% to 27.1%.

This visualization makes it easier to identify geographic patterns and areas that may require further public health investigation. Public health departments could use this type of information to explore where diabetes prevention programs, screening initiatives, and health education activities may be especially important.

3.2 Top 10 U.S. Counties by Estimated Diabetes Prevalence

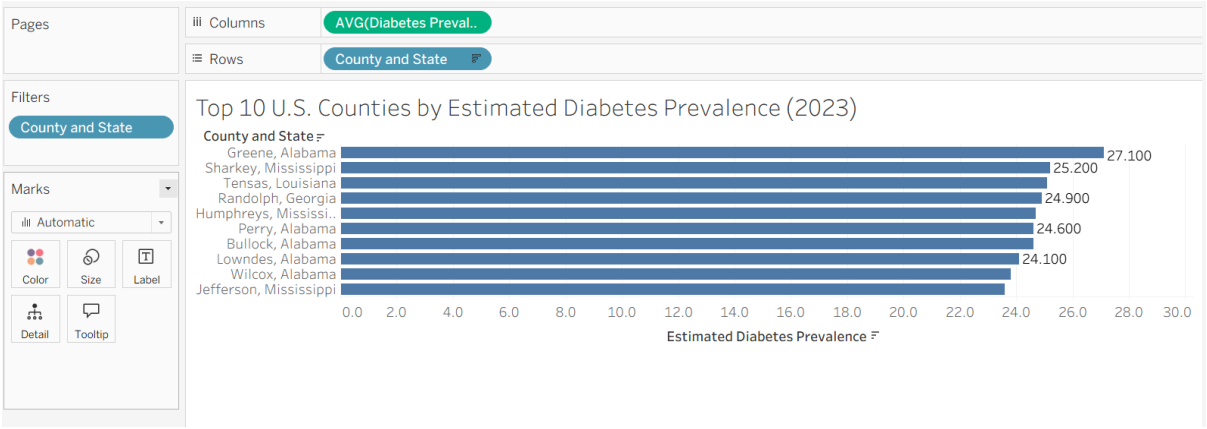


Figure 2. Top 10 U.S. Counties by Estimated Diabetes Prevalence (2023).

The second visualization is a horizontal bar chart showing the ten counties with the highest estimated diabetes prevalence in the dataset. County and state names are displayed along the vertical axis, while average diabetes prevalence is represented along the horizontal axis.

The visualization highlights Greene County, Alabama, with an estimated prevalence of approximately 27.1%. Other counties appearing near the top include Sharkey County, Mississippi, and Tensas Parish, Louisiana. The displayed estimates indicate that these counties have relatively high estimated diabetes prevalence compared with other counties included in the analysis.

The bar chart provides a more focused comparison than the geographic map. While the map helps identify broad geographic patterns, the bar chart makes it easier to compare individual counties and identify areas that may warrant additional investigation.

These results should be interpreted as a comparison of one specific health measure, rather than a ranking of the overall health of these counties. The CDC notes that PLACES estimates are modeled and should not be used to rank the overall health of geographic areas (CDC, 2025c).

Geographic differences in diabetes prevalence have also been reported in previous research. Onufrak et al. (2024) found that diabetes prevalence varied across metropolitan and nonmetropolitan counties and differed by U.S. region. Their findings highlight the importance of considering geographic and socioeconomic characteristics when interpreting county-level health estimates.

3.3 Relationship Between Diabetes and Obesity Prevalence

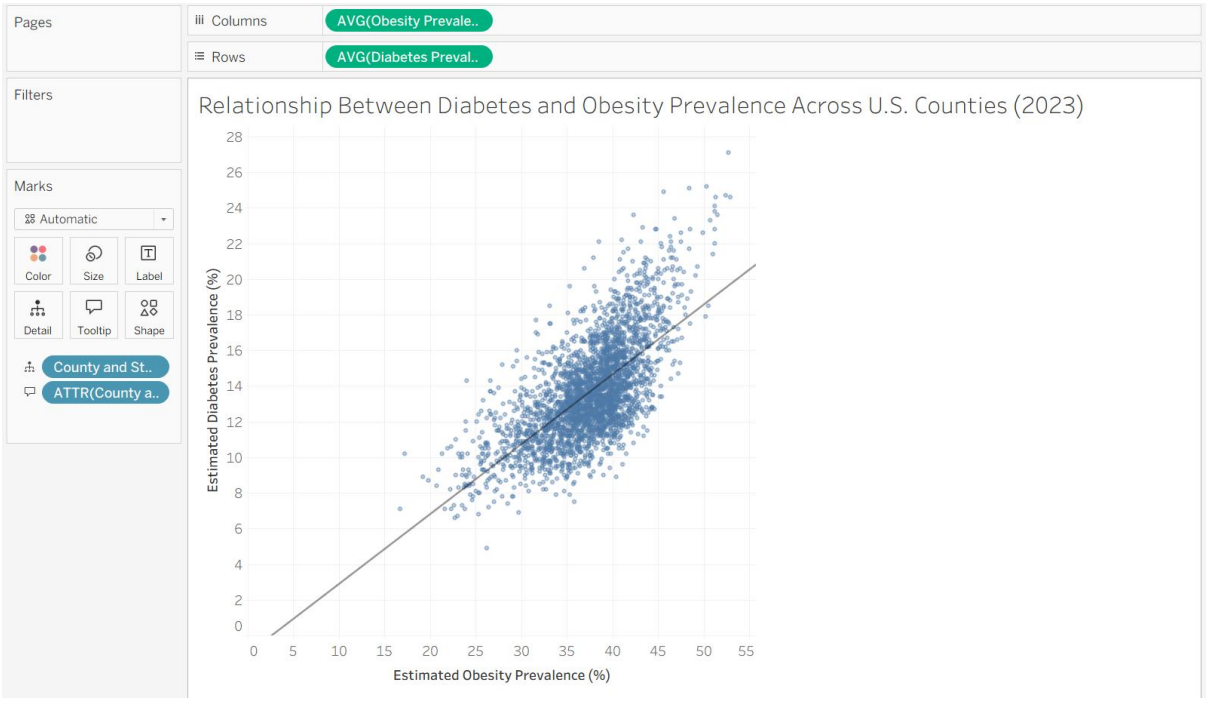


Figure 3. Relationship Between Diabetes and Obesity Prevalence Across U.S. Counties (2023).

The third visualization is a scatter plot examining the relationship between estimated obesity prevalence and estimated diabetes prevalence across U.S. counties. Average obesity prevalence is displayed on the horizontal axis, while average diabetes prevalence is displayed on the vertical axis. Each point represents a county, and a trend line was added to help illustrate the overall pattern.

The scatter plot shows a positive association between obesity prevalence and diabetes prevalence. In general, counties with higher estimated obesity prevalence also tend to have higher estimated diabetes prevalence. The upward direction of the trend line supports this visual observation.

This observed relationship is consistent with previous research identifying obesity as an important risk factor for type 2 diabetes. The National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK, n.d.) explains that overweight and obesity increase the risk of developing type 2 diabetes. However, the relationship between these conditions is also influenced by other biological, behavioral, and environmental factors.

This pattern is relevant to public health because obesity is an important risk factor associated with type 2 diabetes. The visualization suggests that counties with higher levels of obesity may also experience a greater burden of diabetes.

However, the scatter plot demonstrates an association, not a causal relationship. It does not establish that obesity alone explains differences in diabetes prevalence. Other factors, including age distribution, socioeconomic conditions, healthcare access, lifestyle, and other health-related factors, may also contribute to the observed patterns.

3.4 Final Interactive Dashboard

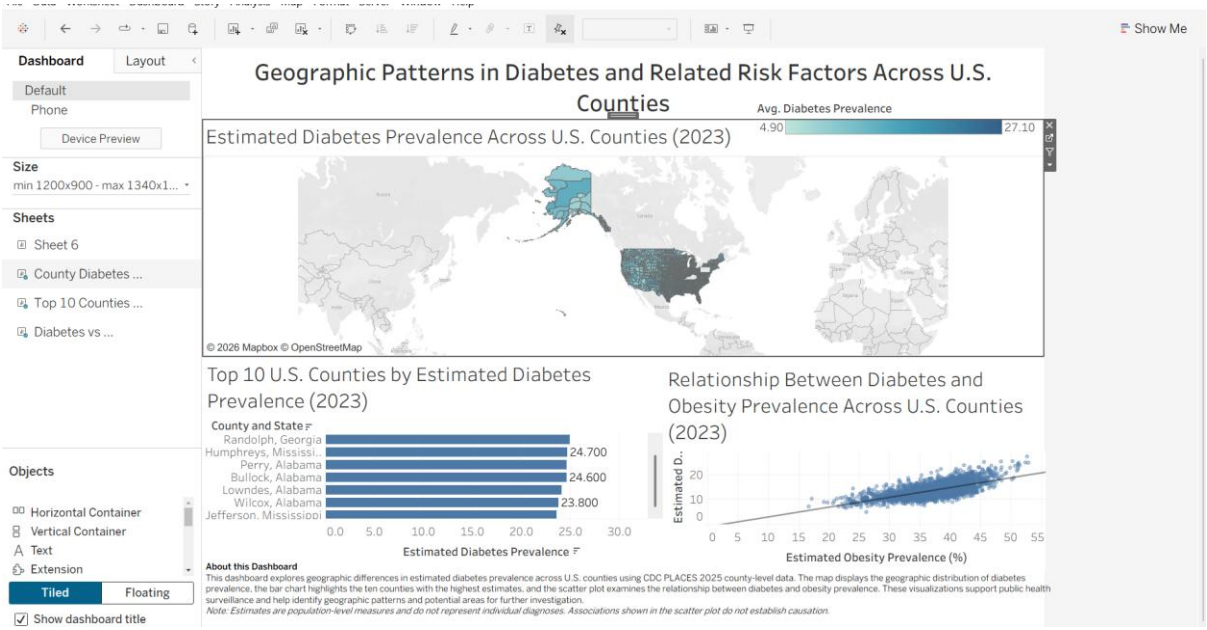


Figure 4. Final Dashboard: Geographic Patterns in Diabetes and Related Risk Factors Across U.S. Counties.

The final dashboard combines the three visualizations into a single report. It presents the geographic distribution of diabetes prevalence, highlights the ten counties with the highest estimated prevalence, and illustrates the relationship between diabetes and obesity prevalence. Combining these visualizations provides a comprehensive overview of the analysis and makes it easier for viewers to compare geographic patterns and interpret the findings in one place.

4. CHALLENGES AND SOLUTIONS

Several challenges were encountered while developing the visualizations in Tableau Public.

One challenge was connecting the dataset to Tableau Public. Since the free version did not provide the PostgreSQL connection option needed for my original database workflow, I used a CSV file as the data source for the analysis. This allowed me to import and explore the data directly in Tableau.

Another challenge involved selecting the appropriate geographic fields and measures when creating the county-level map. Tableau initially displayed an incorrect aggregation for the diabetes prevalence measure. I addressed this by using the average aggregation and checking the map's colour legend to ensure that the visualization represented prevalence values appropriately.

Creating the scatter plot also required adjusting the dimensions of the chart. Initially, the visualization displayed only one aggregated mark. I added the county-level geographic dimension to the Marks card to display individual county observations. This allowed the scatter plot to show the relationship between obesity and diabetes prevalence across counties.

Finally, arranging multiple worksheets on a single dashboard required adjustments to the dashboard size, chart placement, and text formatting. These changes improved the readability and organization of the final report.

5. PUBLIC HEALTH APPLICATIONS

The visualizations developed in this project demonstrate how county-level health data can support public health surveillance and decision-making.

The geographic map can help public health professionals identify areas with higher estimated diabetes prevalence and investigate potential geographic disparities. The bar chart provides a focused comparison of counties with relatively high estimates, helping identify locations for further assessment. The scatter plot provides insight into the relationship between obesity and diabetes prevalence and may support the development of prevention strategies addressing multiple chronic disease risk factors.

For example, local health departments could use these findings to guide further assessment of community needs, plan diabetes awareness campaigns, promote healthy eating and physical activity, and consider where additional screening or preventive health services may be beneficial.

However, these estimates should be used as a starting point for further investigation rather than as the sole basis for allocating healthcare resources. Additional information about population characteristics, socioeconomic conditions, healthcare access, and local community needs would be necessary for more comprehensive planning.

6. LIMITATIONS

This analysis has several limitations. First, the PLACES dataset provides modeled estimates rather than direct measurements of every individual in each county. Second, the analysis focuses on county-level data, so the findings cannot be used to make conclusions about individual patients. Third, the scatter plot identifies an association between diabetes and obesity prevalence but does not establish causation.

Additionally, the analysis focuses on the 2023 estimates included in the 2025 PLACES release. It does not examine changes over multiple years or establish trends over time. Differences in estimated prevalence may also reflect demographic, socioeconomic, and other contextual factors that were not examined in these visualizations.

7. CONCLUSION

This project used Tableau Public to explore geographic patterns in estimated diabetes prevalence and its relationship with obesity across U.S. counties. Three visualizations were developed: a county-level geographic map, a horizontal bar chart of the ten counties with the highest estimated diabetes prevalence, and a scatter plot examining diabetes and obesity prevalence.

The visualizations demonstrate geographic variation in diabetes prevalence and a positive association between diabetes and obesity estimates. Together, they show how public health data can be transformed into visual information that is easier to interpret and communicate.

Through this project, I gained practical experience in importing and preparing public health data, selecting appropriate visualizations, working with geographic information in Tableau, and interpreting population-level health indicators. Overall, the project demonstrates the value of data visualization as a tool for public health surveillance, identifying areas for further investigation, and supporting evidence-informed healthcare planning.

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